



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to GOWSALYA AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

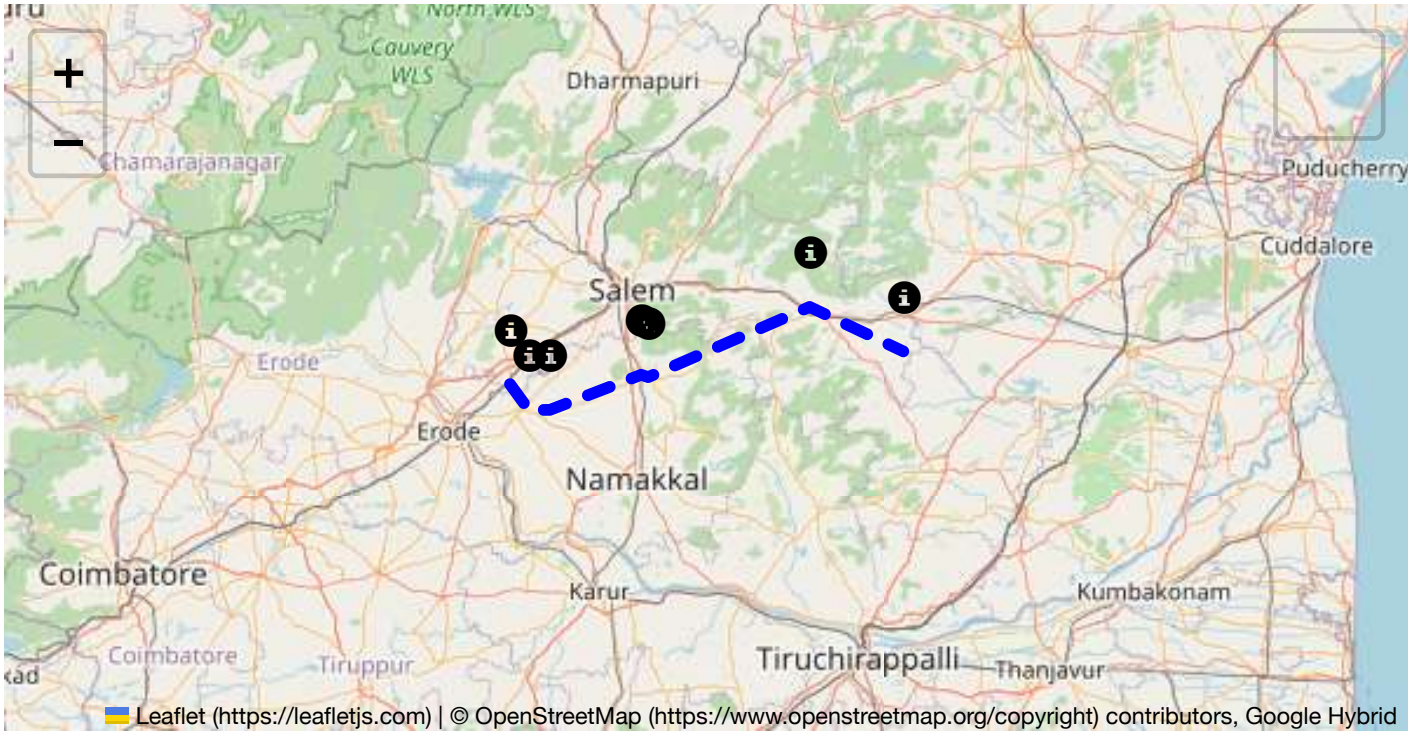
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

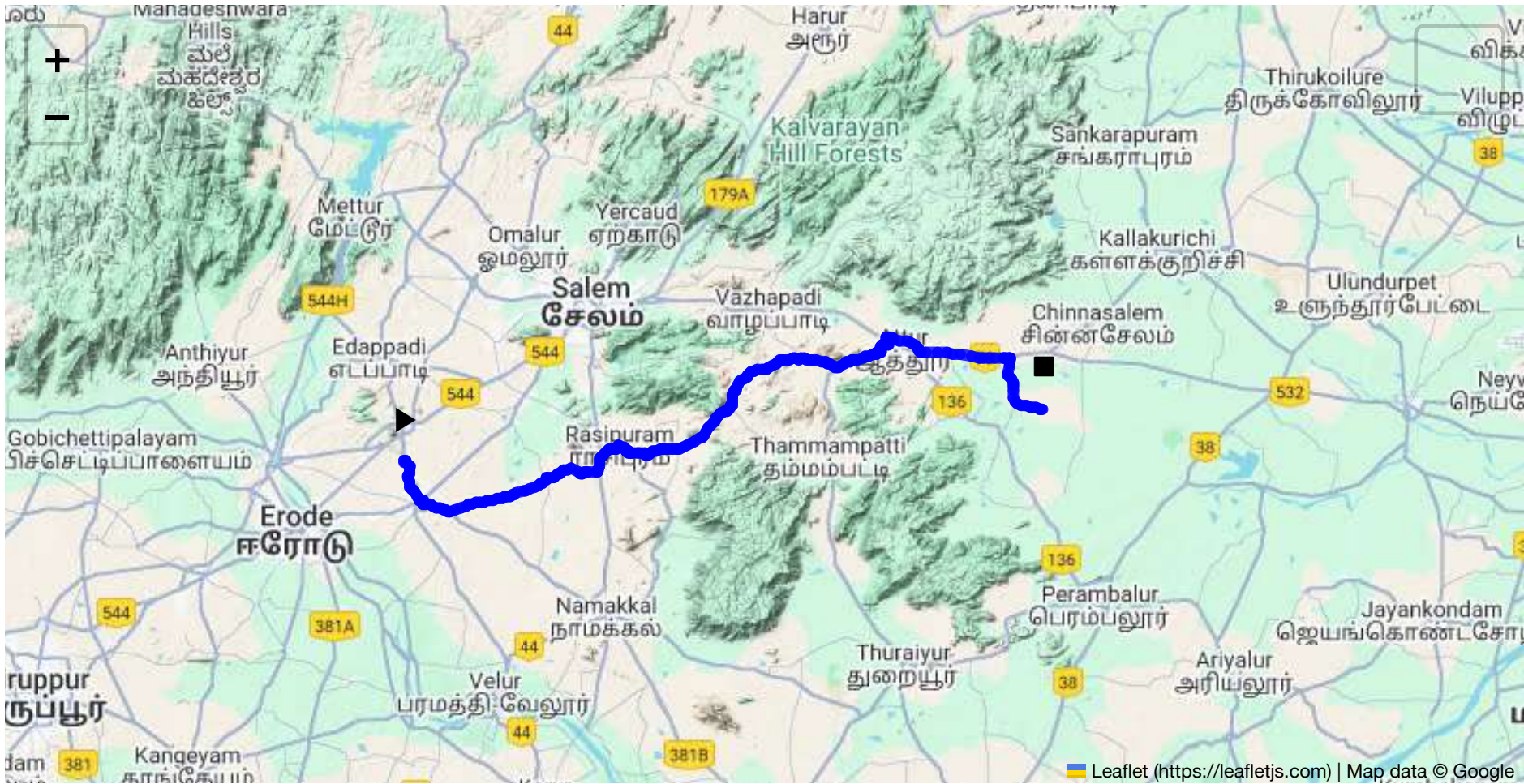
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 131.68 km**  
**Estimated Duration: 3.0 hours**  
**Adjusted Duration (Heavy Vehicle): 3.7 hours**

Start: (11.4381, 77.8734)  
End: (11.51324, 78.79597)



## Welcome to the Journey Risk Management Study

- Overview of the Route Map:**
  - The route from CVQF+23W, Sangagiri, to GQ7W+79, Puliyankurichi, passes through key waypoints, including Namakkal, SH 79, Kavundampalaiyam, Rasipuram, and Koneripatti. The journey spans approximately 131.68 kilometers.
- Typical Weather Conditions and Potential Weather-Related Hazards:**
  - Tamil Nadu often experiences hot, humid weather. During monsoon (June to September), heavy rains can cause flooding and poor road visibility. Cyclonic activity, albeit rare, can affect the coastal regions and occasionally impact inland weather, causing heavy winds and rainfall.
- Analysis of Traffic Patterns:**
  - Peak traffic hours are typically from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM, especially in urban areas like Namakkal and Rasipuram. Congestion is common near city centers and markets, particularly during festive seasons.
- Assessment of Road Quality and Infrastructure:**
  - State highways like SH 79 generally maintain good conditions but can present potholes and uneven surfaces, particularly after the monsoon season. Some stretches may lack proper lighting, affecting night-time visibility.
- Suggestions for Alternative Routes for Emergencies:**
  - If an alternate route is necessary, consider using NH 44 for bigger sections, as it offers better facilities and connectivity. State highways SH 95 or SH 86 can serve as detours, depending on the specific emergency location.
- Summary of Local Regulations Affecting Hazardous Material Transport:**
  - Transporting hazardous materials requires special permits. Regulations mandate specific labeling and precautions for additional safety. There are designated routes for such materials,

often avoiding highly populated areas.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:
- Past incidents often involve driver fatigue or mechanical failure. While rare, mishaps could lead to road closures and require adherence to safety protocols and emergency management plans.
8. Environmental Considerations and Sensitive Areas:
- The route traverses agricultural zones and rural landscapes. Drivers should be aware of wildlife crossing areas and avoid spillage or littering to prevent environmental harm.
9. Analysis of Communication Coverage:
- Communication coverage is generally reliable in urban areas, though rural stretches might experience brief connectivity issues. Major providers cover the region, but it is advisable to have alternate means of communication.
10. Estimated Emergency Response Times:
- In urban areas, emergency response could take 15-30 minutes. In rural stretches or during peak traffic, response times might extend to one hour or more.
11. Overall Summary of Risk Assessment:
- The route presents challenges typical of regional travel in India, including variable road quality and congestion. Weather and emergency management are critical considerations due to potential for flooding and extended response times. Drivers should be well-versed in local regulations for hazardous materials and environmental protection measures. Proper route planning and adherence to safety protocols can minimize risks associated with this transit path.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates            | Speed Limit |
|----|------------|------------|------------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480     | 15 KM/Hr    |
| 1  | U-Turn     | High       | 11.4535263, 78.1964113 | 10 KM/Hr    |
| 2  | Turn       | High       | 11.43968, 77.87345     | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44019, 77.87547     | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.39685, 77.88861     | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.38376, 77.89914     | 30 KM/Hr    |
| 6  | Turn       | High       | 11.38186, 77.90258     | 15 KM/Hr    |
| 7  | Turn       | High       | 11.37828, 77.90171     | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.36754, 77.93919     | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.41354, 78.08075     | 30 KM/Hr    |
| 10 | Turn       | High       | 11.41447, 78.08063     | 15 KM/Hr    |
| 11 | Turn       | High       | 11.42964, 78.11587     | 15 KM/Hr    |
| 12 | Turn       | Medium     | 11.42447, 78.15264     | 30 KM/Hr    |
| 13 | Turn       | High       | 11.42611, 78.15532     | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 14 | Turn       | High       | 11.44521, 78.15833 | 15 KM/Hr    |
| 15 | Turn       | Medium     | 11.45566, 78.17361 | 30 KM/Hr    |
| 16 | Turn       | High       | 11.45587, 78.18057 | 15 KM/Hr    |
| 17 | Blind Spot | Blind Spot | 11.46035, 78.18534 | 10 KM/Hr    |
| 18 | Turn       | High       | 11.45929, 78.18544 | 15 KM/Hr    |
| 19 | Turn       | Medium     | 11.45764, 78.19208 | 30 KM/Hr    |
| 20 | Blind Spot | Blind Spot | 11.45084, 78.20378 | 10 KM/Hr    |
| 21 | Blind Spot | Blind Spot | 11.45354, 78.19642 | 10 KM/Hr    |
| 22 | Turn       | Medium     | 11.45117, 78.20024 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.44993, 78.22152 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.44896, 78.22294 | 30 KM/Hr    |
| 25 | Turn       | High       | 11.44797, 78.22314 | 15 KM/Hr    |
| 26 | Turn       | Medium     | 11.44756, 78.22414 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.45353, 78.23584 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.49835, 78.32417 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.50784, 78.33331 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.50865, 78.33966 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.51183, 78.34416 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 11.51870, 78.34816 | 30 KM/Hr    |
| 33 | Turn       | Medium     | 11.54842, 78.35627 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.56719, 78.37813 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 11.56940, 78.39856 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 11.57497, 78.49389 | 30 KM/Hr    |
| 37 | Turn       | High       | 11.57093, 78.49993 | 15 KM/Hr    |
| 38 | Turn       | Medium     | 11.57829, 78.51245 | 30 KM/Hr    |
| 39 | Turn       | Medium     | 11.59108, 78.55637 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 11.59253, 78.55778 | 30 KM/Hr    |
| 41 | Turn       | High       | 11.59333, 78.56209 | 15 KM/Hr    |
| 42 | Turn       | High       | 11.60406, 78.56494 | 15 KM/Hr    |
| 43 | Turn       | Medium     | 11.60471, 78.57359 | 30 KM/Hr    |
| 44 | Turn       | High       | 11.60491, 78.57377 | 15 KM/Hr    |
| 45 | Turn       | Medium     | 11.61294, 78.57255 | 30 KM/Hr    |
| 46 | Turn       | High       | 11.61403, 78.57272 | 15 KM/Hr    |
| 47 | Turn       | Medium     | 11.59279, 78.62141 | 30 KM/Hr    |
| 48 | Turn       | Medium     | 11.58562, 78.74231 | 30 KM/Hr    |
| 49 | Turn       | Medium     | 11.58530, 78.74272 | 30 KM/Hr    |
| 50 | Turn       | High       | 11.58553, 78.74952 | 15 KM/Hr    |
| 51 | Turn       | High       | 11.52182, 78.75791 | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 52 | Turn      | Medium     | 11.51306, 78.79502 | 30 KM/Hr    |
| 53 | Turn      | High       | 11.51331, 78.79585 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                                      | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital          | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode           | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | Jayaa Hospital                            | 11.3848223, 77.9021886 | 30 km/h     | Medium     |
| 14 | hospital | Sudha Hospital                            | 11.4555773, 78.1760125 | 30 km/h     | Medium     |
| 15 | hospital | Government Hospital Blood Bank, Rasipuram | 11.4612082, 78.1849249 | 30 km/h     | Medium     |
| 16 | hospital | M.R. Hospital, Namakkal                   | 11.4613462, 78.1861702 | 30 km/h     | Medium     |
| 17 | hospital | dr.thangavel dch hospital                 | 11.4619955, 78.1849181 | 30 km/h     | Medium     |
| 18 | police   | Police Station Rasipuram                  | 11.4617825, 78.1847411 | 30 km/h     | Medium     |
| 19 | hospital | surya hospital                            | 11.4622163, 78.1840544 | 30 km/h     | Medium     |
| 20 | hospital | Government Hospital, Rasipuram            | 11.4612987, 78.1851369 | 30 km/h     | Medium     |
| 21 | hospital | Rasi Hospital                             | 11.4618368, 78.1855695 | 30 km/h     | Medium     |
| 22 | hospital | R K Eye Care Center                       | 11.4589079, 78.1913596 | 30 km/h     | Medium     |
| 23 | hospital | Manickkam Hospital                        | 11.4568944, 78.1926061 | 30 km/h     | Medium     |
| 24 | hospital | Snathi Hospital                           | 11.4564245, 78.1932485 | 30 km/h     | Medium     |
| 26 | hospital | Sugam Hospital,Rasipuram                  | 11.4553007, 78.1936221 | 30 km/h     | Medium     |
| 27 | hospital | Government Hospital, Namagiripettai       | 11.4569484, 78.265263  | 30 km/h     | Medium     |
| 28 | clinic   | Dr. Saravanakumar Clinic                  | 11.5842002, 78.4441223 | 30 km/h     | Medium     |

|    | type     | name                  | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------|---------------------------|-------------|------------|
| 29 | hospital | Government Hospital   | 11.572472, 78.497407      | 30 km/h     | Medium     |
| 31 | hospital | Abi Hospital          | 11.59319, 78.560129       | 30 km/h     | Medium     |
| 34 | clinic   | Sugam Clinic          | 11.5832481,<br>78.7496438 | 30 km/h     | Medium     |
| 35 | hospital | Selvam Hospital       | 11.585978, 78.748761      | 30 km/h     | Medium     |
| 36 | hospital | Prabushankar Hospital | 11.584239,<br>78.7490323  | 30 km/h     | Medium     |

### Crowded Spots

|    | type        | name                                             | coordinates               | speed_limit | risk_level |
|----|-------------|--------------------------------------------------|---------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி                     | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி                    | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை              | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 6  | marketplace | Farmer Market                                    | 11.4146849,<br>78.0805463 | 30 km/h     | Medium     |
| 7  | school      | Government Higher Secondary School               | 11.4150045,<br>78.0838172 | 30 km/h     | Medium     |
| 8  | school      | Government school                                | 11.4248027,<br>78.1323902 | 30 km/h     | Medium     |
| 9  | college     | Paavai Arts And Science College For Women        | 11.4262443,<br>78.1546987 | 30 km/h     | Medium     |
| 10 | school      | SRV EXCEL Matriculation Higher Secondary School  | 11.4299679,<br>78.152616  | 30 km/h     | Medium     |
| 11 | college     | Thiruvalluvar Govt arts college                  | 11.4449352,<br>78.1595562 | 30 km/h     | Medium     |
| 12 | school      | Venkateswara School                              | 11.4468258,<br>78.1595761 | 30 km/h     | Medium     |
| 13 | school      | Venkateswara Middle school                       | 11.4472063,<br>78.1596191 | 30 km/h     | Medium     |
| 25 | school      | Govt. Hr. Sec. School                            | 11.4585324,<br>78.1933666 | 30 km/h     | Medium     |
| 30 | school      | Rasi Matric HR Sec School                        | 11.5760027,<br>78.5062808 | 30 km/h     | Medium     |
| 32 | school      | MARUTHI PRIMAR, SECONDAR ,HIGHER SECONDARY SHOOL | 11.5859471,<br>78.7186322 | 30 km/h     | Medium     |

|    | type    | name                        | coordinates               | speed_limit | risk_level |
|----|---------|-----------------------------|---------------------------|-------------|------------|
| 33 | college | Maruthi Polytechnic College | 11.5856852,<br>78.7201072 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.38395, 77.89480



Risk Type: U-Turn  
Risk Level: High  
Speed Limit: 10 KM/Hr  
Coordinates: 11.4535263, 78.1964113



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44019, 77.87547



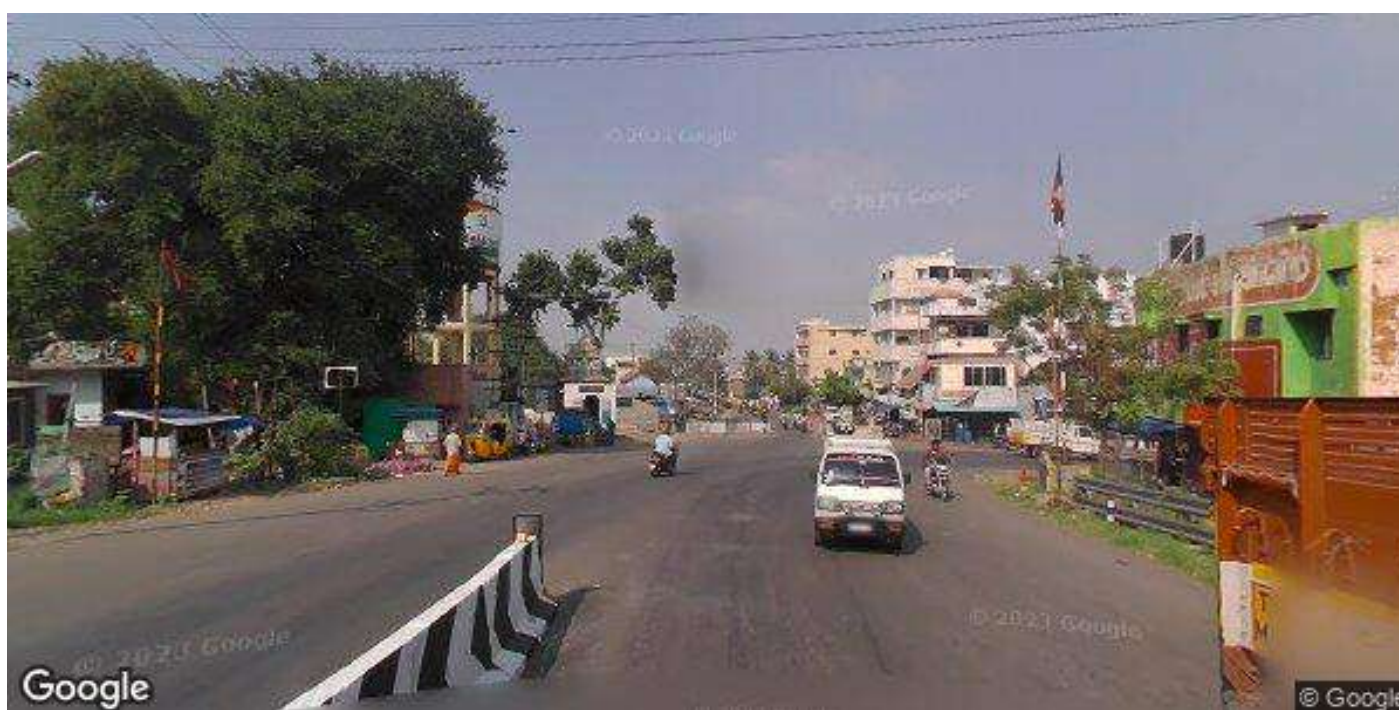
**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.39685, 77.88861



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.38376, 77.89914



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.38186, 77.90258



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.37828, 77.90171



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.36754, 77.93919



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.41354, 78.08075



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.41447, 78.08063



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.42964, 78.11587



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.42447, 78.15264



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.42611, 78.15532



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44521, 78.15833



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45566, 78.17361



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45587, 78.18057



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.46035, 78.18534



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45929, 78.18544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45764, 78.19208



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45084, 78.20378



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45354, 78.19642



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45117, 78.20024



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44993, 78.22152



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44896, 78.22294



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44797, 78.22314



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44756, 78.22414



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45353, 78.23584



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49835, 78.32417



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50784, 78.33331



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50865, 78.33966



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.51183, 78.34416



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.51870, 78.34816



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.54842, 78.35627



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56719, 78.37813



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56940, 78.39856



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57497, 78.49389



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.57093, 78.49993



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57829, 78.51245



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.59333, 78.56209



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.60406, 78.56494



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.60471, 78.57359



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.60491, 78.57377



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.61294, 78.57255



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.61403, 78.57272



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.59279, 78.62141



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.58562, 78.74231



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.58530, 78.74272



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.58553, 78.74952



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.52182, 78.75791

